

FOOD TESTS ON CARBOHYDRATES

1. Test for reducing sugars

The reagent used is

Benedict's solution (blue)
or Fehling's solution (blue).

Boiling is required.

Reducing sugar test

Procedure	Observation	Conclusion
To 1 cm³ of food solution, add 1 cm³ of Benedict's solution and boil.	Colourless or turbid solution turned to a blue solution , then to a green solution , to a yellow precipitate , to orange precipitate and to a brown precipitate on boiling.	<i>Little or Moderate or Much or Too much; reducing sugars present.</i>
	Colourless or turbid solution turned to a blue solution which persists on boiling.	<i>Reducing sugars absent.</i>

If **Fehling's solution** is used, the change is from blue solution to orange precipitate if reducing sugars are present. It remains a blue solution if they are absent.

Examples of reducing sugars include:

- ❖ **Glucose** (*present in grapes*)
- ❖ **Fructose** (*present in many edible fruits*)
- ❖ **Galactose** (*present in milk*)
- ❖ **Maltose** (*present in germinating seeds*)

Benedict's Test

<https://www.youtube.com/watch?v=MRAVhZA77Q>

Test for non reducing sugars

procedure	Observation	conclusion
<i>To 1 cm³ of food solution add 1 cm³ of dilute hydrochloric acid and boil, cool under water then add 1 cm³ of sodium hydroxide solution, followed by 1 cm³ of Benedict's solution and boil.</i>	Colourless or turbid solution turned to a <i>blue solution</i> , then to a <i>green solution</i> , to a <i>yellow precipitate</i> and to a <i>brown precipitate</i> on boiling.	Little or Moderate or Much or Too much ; non-reducing sugars present.
	Colourless or turbid solution turned to a <i>blue solution</i> which <i>persists on boiling</i> .	Non-reducing sugars absent .

Test for Non reducing sugars

<https://www.youtube.com/watch?v=QU0VBcHnQOk>

Note:

- ❖ When boiled with **dilute HCl**, the non-reducing sugars **breaks down** into the **reducing sugars**.
- ❖ Sodium hydroxide solution or sodium hydrogen carbonate powder is added to neutralize the acid so that Benedict's solution can work.

Examples of non-reducing sugars include:

1. Sucrose (present in sugar cane)
2. Lactose (present in milk)

Test for starch (IODINE TEST)

Procedure	Observation	Conclusion
To 1 cm ³ of food solution, add 3 drops of iodine solution.	Colourless or turbid solution turned to a black or blue-black or blue solution or brown solution with <i>black specks</i> .	Much or moderate or little starch present .
	Colourless or turbid solution turned to a yellow or brown solution.	Starch absent .

Test for starch

https://www.youtube.com/watch?v=EBbiQw7Jol_k

Starch in different food samples

<https://www.youtube.com/watch?v=5YMDOX71Tel>